

THE OXFORD
SOLID STATE
BASICS

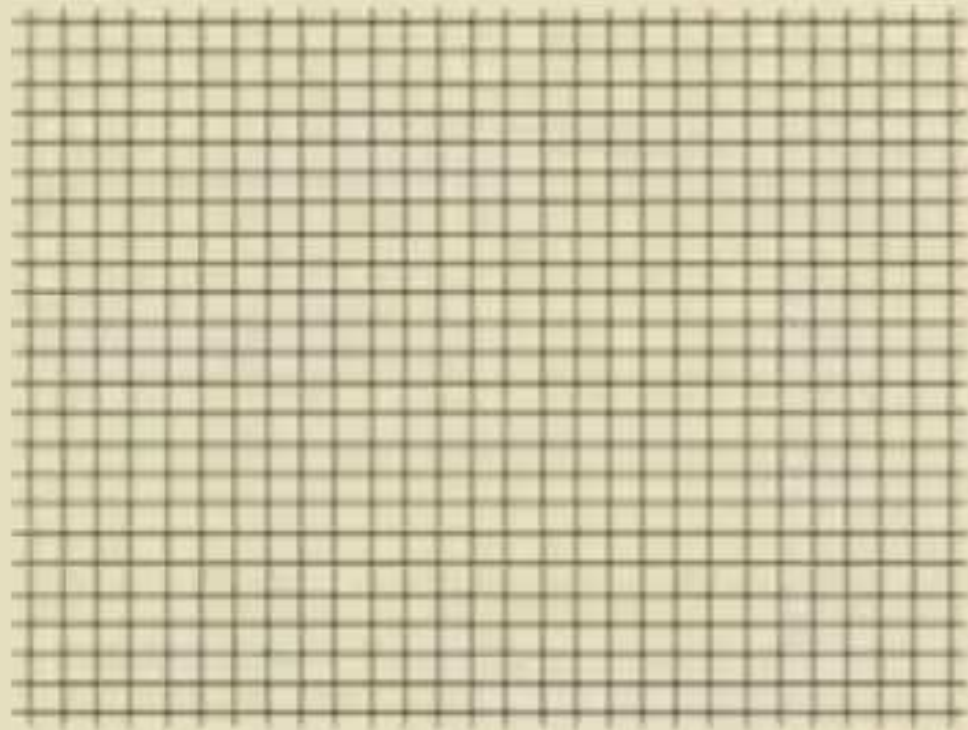


Slides

Condensed Matter Physics

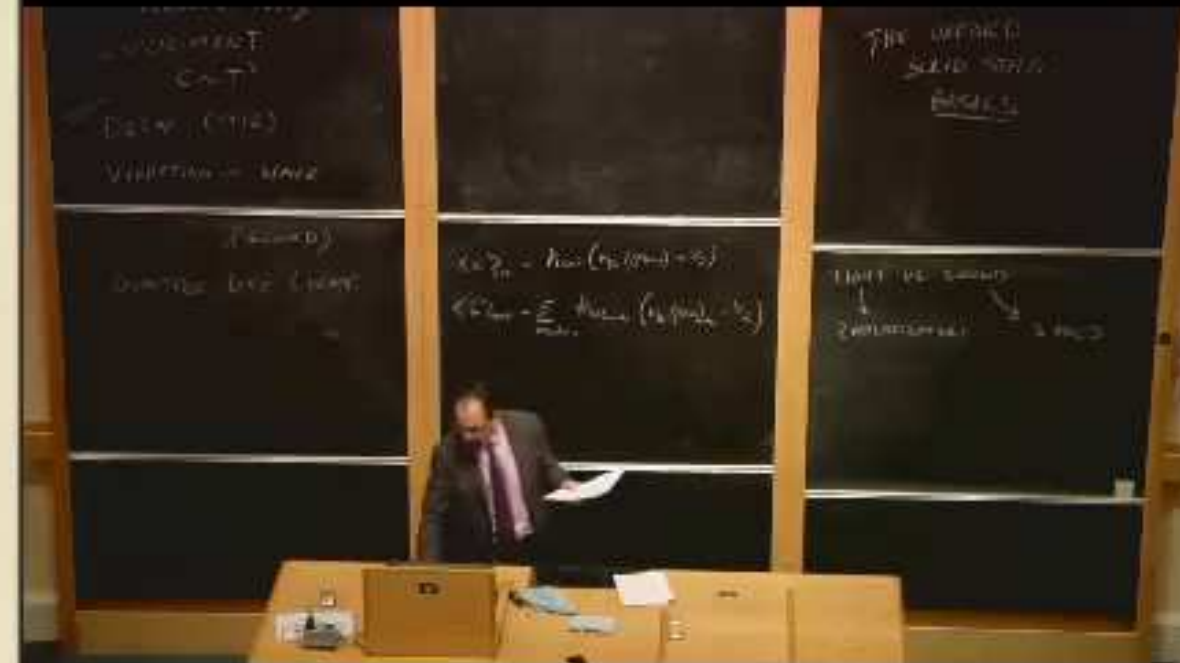
Lecture 2





Longitudinal Wave

http://en.wikipedia.org/wiki/Longitudinal_wave



EXPERIMENT
 $C \sim T^3$

DERIVE (1912)

VIBRATION = WAVE

(SOUND)

QUANTIZE LIKE LIGHT.

$$\langle E \rangle_{\text{osc}} = \hbar \omega \left(n_B(\beta \hbar \omega) + \frac{1}{2} \right)$$

$$\langle E \rangle_{\text{TOT}} = \sum_{\text{modes}} \hbar \omega_{\text{mode}} \left(n_B(\beta \hbar \omega_{\text{mode}}) + \frac{1}{2} \right)$$

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LIGHT VS SOUND

↓
2 POLARIZATIONS

↓
3 POL'S

Molar Specific Heat of Silver

